SHAHEER ALI
25k-3056
BSE-2B

LAB-05 TASKS

IMPORTANT NOTE: Sir my laptop has drivers issue so whole display is distorted and is not straight that's why my output on vs code has issues, sorry for that. And code is actually fine because I copied that instead of pasting screenshot.

TASK-1 (Code)

```
class Student {
    static int totalStudents = 0;
    private int ID;
    private String name;

    //Setters and Getters
    public void setID(int ID) {
        this.ID = ID;
    }
    public int getID() {
        return ID;
    }
    public void setName(String name) {
        this.name = name;
    }
    public String getName() {
        return name;
    }

    //Constructor
    public Student(int ID, String name) {
        this.ID = ID;
        this.name = name;
        Student.totalStudents++;
    }

    //Static Method
    public static void displayTotal() {
        System.out.println("\nTotal no. of students: " +
totalStudents);
```

SHAHEER ALI
25k-3056
BSE-2B

```
    }

    //Method to display details
    public void displayInfo() {
        System.out.println("==== Student Details ====");
        System.out.println("ID : " + ID);
        System.out.println("Name : " + name);
    }
}

class task1 {
    public static void main(String[] args) {
        Student s1 = new Student(123, "Ali");
        Student s2 = new Student(456, "Sherry");
        Student s3 = new Student(789, "Owais");


        s1.displayInfo();
        s2.displayInfo();
        s3.displayInfo();

        Student.displayTotal();
    }
}
```

Output (Task-1):

```
==== Student Details ====
ID : 123
Name : Ali
==== Student Details ====
ID : 456
Name : Sherry
==== Student Details ====
ID : 789
Name : Owais

Total no. of Students: 3
```

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TASK-2 (Code)

WITH COMPOSITION:

```
class Laptop {

    private String brandName;
    private int ram;  //in GB
    private int batteryCapacity; //in mAh

    //Setters and Getters
    public void setBrandName(String brandName) {
        this.brandName = brandName;
    }
    public String getBrandName() {
        return brandName;
    }
    public void setRam(int ram) {
        this.ram = ram;
    }
    public int getRam() {
        return ram;
    }
    public void setBatteryCapacity(int batteryCapacity) {
        this.batteryCapacity = batteryCapacity;
    }
    public int getBatteryCapacity() {
        return batteryCapacity;
    }

    //Constructor
    public Laptop(String brandName, int ram, int batteryCapacity) {
        this.brandName = brandName;
        this.ram = ram;
        this.batteryCapacity = batteryCapacity;
    }

    //Methods
    public double calculateBatteryBackup() {
        return batteryCapacity / (ram * 500.0);
```

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25k-3056
BSE-2B

```
    }
}

class Student {
    private int ID;
    private String name;
    private Laptop laptop;

    //Setters and Getters
    public void setID(int ID) {
        this.ID = ID;
    }
    public int getID() {
        return ID;
    }
    public void setName(String name) {
        this.name = name;
    }
    public String getName() {
        return name;
    }

    //Constructor
    public Student(int ID, String name, String brandName, int ram, int
batteryCapacity) {
        this.ID = ID;
        this.name = name;
        laptop = new Laptop(brandName, ram, batteryCapacity);
    }

    //Method
    public void displayInfo() {
        System.out.println("==== Student Information ====");
        System.out.println("ID : " + ID);
        System.out.println("Name : " + name);
        System.out.println("----Laptop Info----");
        System.out.println("Brand name : " + laptop.getBrandName());
        System.out.println("Ram : " + laptop.getRam());
```

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25k-3056
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```
        System.out.println("Battery Capacity : " +
laptop.getBatteryCapacity());
        System.out.println("Battery Backup : " +
laptop.calculateBatteryBackup() + " hours");
    }
}

public class task2 {
   public static void main(String[] args) {
       Student s1 = new Student(456, "Sherry", "Dell", 8, 4500);
       Student s2 = new Student(123, "Ali", "Lenovo", 16, 5000);

       s1.displayInfo();
       s2.displayInfo();
   }
}
```

WITH AGGREGATION:

```
class Laptop {

    private String brandName;
    private int ram;  //in GB
    private int batteryCapacity; //in mAh

    //Setters and Getters
    public void setBrandName(String brandName) {
        this.brandName = brandName;
    }
    public String getBrandName() {
        return brandName;
    }
    public void setRam(int ram) {
        this.ram = ram;
    }
    public int getRam() {
```

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25k-3056
BSE-2B

```
        return ram;
    }
    public void setBatteryCapacity(int batteryCapacity) {
        this.batteryCapacity = batteryCapacity;
    }
    public int getBatteryCapacity() {
        return batteryCapacity;
    }

    //Constructor
    public Laptop(String brandName, int ram, int batteryCapacity) {
        this.brandName = brandName;
        this.ram = ram;
        this.batteryCapacity = batteryCapacity;
    }

    //Methods
    public double calculateBatteryBackup() {
        return batteryCapacity / (ram * 500.0);
    }
}

class Student {
    private int ID;
    private String name;
    private Laptop laptop;

    //Setters and Getters
    public void setID(int ID) {
        this.ID = ID;
    }
    public int getID() {
        return ID;
    }
    public void setName(String name) {
        this.name = name;
    }
    public String getName() {
        return name;
```

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BSE-2B

```
    }

    //Constructor
    public Student(int ID, String name, Laptop laptop) {
        this.ID = ID;
        this.name = name;
        this.laptop = laptop;
    }

    //Method
    public void displayInfo() {
        System.out.println("==== Student Information ====");
        System.out.println("ID : " + ID);
        System.out.println("Name : " + name);
        System.out.println("----Laptop Info----");
        System.out.println("Brand name : " + laptop.getBrandName());
        System.out.println("Ram : " + laptop.getRam());
        System.out.println("Battery Capacity : " +
laptop.getBatteryCapacity());
        System.out.println("Battery Backup : " +
laptop.calculateBatteryBackup() + " hours");
    }
}

public class task2 {
   public static void main(String[] args) {
       Laptop l1 = new Laptop("Dell", 8, 4500);
       Laptop l2 = new Laptop("Lenovo", 16, 5000);
        //Laptop objects created in main and passed as parameters
(aggregation)
        //So their lifecycle is independent of Student class
       Student s1 = new Student(456, "Sherry", l1);
       Student s2 = new Student(123, "Ali", l2);

       s1.displayInfo();
       s2.displayInfo();
   }
}
```

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Output (Task-2):

```
==== Student Information ====
ID : 456
Name : Sherry
----Laptop Info----
Brand name : Dell
Ram : 8
Battery Capacity : 4500
Battery Backup : 1.125 hours
==== Student Information ====
ID : 123
Name : Ali
----Laptop Info----
Brand name : Lenovo
Ram : 16
Battery Capacity : 5000
Battery Backup : 0.625 hours
```

Sir sorry the distortion is due to display drivers issue…..

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TASK-3 (Code):

```
class Employee {
    private int ID;
    private String name;
    private double basicSalary;
    //Static fields shared by all object:
    private static double bonusPool = 0; //global/total bonus pool
    private static double bonusPercentage = 10; //assumed fixed
    private static int totalEmployees = 0;


    //Constructor
    public Employee(int ID, String name, double basicSalary) {
        this.ID = ID;
        this.name = name;
        this.basicSalary = basicSalary;
        bonusPool = bonusPool + (basicSalary * bonusPercentage / 100);
        totalEmployees++;
    }

    //Getters
    public int getId() {
        return ID;
    }
    public String getName() {
        return name;
    }
    public double getBasicSalary() {
        return basicSalary;
    }

    //Methods
    public double calculateTotalCompensation() {
        return (basicSalary + (bonusPool / totalEmployees));
    }
```

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```
    public static void displayBonusPool() {
        System.out.println("Total Bonus Pool: " + Employee.bonusPool);
    }

    public void displayInfo() {
        System.out.println("==== Employee Details ====");
        System.out.println("ID: " + ID);
        System.out.println("Name: " + name);
        System.out.println("Basic salary: " + basicSalary);
        System.out.println("Total Compensation: " +
calculateTotalCompensation() + "\n");
    }
}

public class task3 {
    public static void main(String[] args) {
        Employee e1 = new Employee(123,"Sherry", 20000);
        Employee e2 = new Employee(456, "Ali", 30000);
        Employee e3 = new Employee(789, "Owais", 35000);
        Employee e4 = new Employee(101, "Shayan", 45000);

        e1.displayInfo();
        e2.displayInfo();
        e3.displayInfo();
        e4.displayInfo();

        Employee.displayBonusPool();
    }
}
```

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Output (Task-3):

```
=== Employee Details ===
ID: 123
Name: Sherry
Basic salary: 20000.0
Total Compensation: 32500.0

=== Employee Details ===
ID: 456
Name: Ali
Basic salary: 30000.0
Total Compensation: 33250.0

=== Employee Details ===
ID: 789
Name: Owais
Basic salary: 35000.0
Total Compensation: 38250.0

=== Employee Details ===
ID: 101
Name: Shawn
Basic salary: 45000.0
Total Compensation: 48250.0

Total Bonus Pool: 15000.0
PS C:\Users\Acquisitator\Documents\Record S
```

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TASK-4 (Code)

```
class Device {
    private int deviceID;
    private double powerRating; //in watts
    private double hoursUsedPerDay; //per day

    //Constructor
    public Device(int deviceID, double powerRating, double
hoursUsedPerDay){
        this.deviceID = deviceID;
        this.powerRating = powerRating;
        this.hoursUsedPerDay = hoursUsedPerDay;
    }

    //Method
    public double calculateDailyEnergyUsage(){
        return (powerRating * hoursUsedPerDay);
    }
}

class Room {
    private Device device; //device object

    //Constructor
    public Room(int deviceID, double powerRating, double
hoursUsedPerDay){
       this.device = new Device(deviceID, powerRating,
hoursUsedPerDay);
    }

    public double energyUsage(){
        return device.calculateDailyEnergyUsage();
    }
}

class House {
    private int ID;
    private String ownerName;
```

SHAHEER ALI
25k-3056
BSE-2B

```java
    private Room room; //room object

    //Constructor
    public House(int ID, String ownerName, int deviceID, double
powerRating, double hoursUsedPerDay){
        this.ID = ID;
        this.ownerName = ownerName;
        this.room = new Room(deviceID, powerRating, hoursUsedPerDay);
    }

    //Method
    public double totalDailyElectricityConsumption(){
        return room.energyUsage();
    }
}


public class task4 {
    public static void main(String[] args) {
        House house = new House(123, "Noman", 321, 756.54, 3);

        System.out.println("Total daily electricity consumption: " +
house.totalDailyElectricityConsumption());
    }
}
```

Output (Task-4):

```
public class task4 {
    Run | Debug
    public static void main(String[] args) {
        House house = new House(ID: 123, ownerName: "Noman", deviceID: 321, pow

PROBLEMS 8   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

Total daily electricity consumption: 2269.62
PS C:\Users\Administrator\Documents\Second Semester\OOP LAB\Lab-05\Lab-05 (Tasks)>
```

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TASK-5 (Code)

```
class Student {
    private int ID;
    private String name;
    private double marks;

    //Getters
    public int getID() {
        return ID;
    }
    public String getName() {
        return name;
    }
    public double getMarks() {
        return marks;
    }

    //Constructor
    public Student(int ID, String name, double marks) {
        this.ID = ID;
        this.name = name;
        this.marks = marks;
    }

    //Methods
    public void displayStudentDetails() {
        System.out.println("==== Student Information ====");
        System.out.println("ID: " + ID);
        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
    }

    public boolean isPassed() {
        if(marks >= 40){
            return true;
        }
        else{
            return false;
```

SHAHEER ALI
25k-3056
BSE-2B

```
        }
    }
}

class College {
    private String collegeName;
    private String city;
    private Student[] students; //Student class array object
    private int studentCount;

    //Constructor
    public College(String collegeName, String city, Student[]
students) {
        this.collegeName = collegeName;
        this.city = city;
        this.students = students;
        this.studentCount = students.length;
    }

    //Method
    public void displayCollegeDetails() {
        System.out.println("==== College Information ====");
        System.out.println("Name: " + collegeName);
        System.out.println("City: " + city);

        System.out.println("\n==== Student Information ====");

        if(students == null || studentCount == 0) {
            System.out.println("No students enrolled!");
            return;
        }
        for(int i=0; i<studentCount; i++){
            students[i].displayStudentDetails();
        }

        int p=0, f=0;
        double sum = 0;
        for(int i=0; i<studentCount; i++){
           sum += students[i].getMarks();
```

SHAHEER ALI
25k-3056
BSE-2B

```
            if(students[i].isPassed()){
             p++;
            }
            else{
             f++;
            }
        }
        System.out.println("\nTotal Students: " + studentCount);
        System.out.println("Passed Students: " + p);
        System.out.println("Failed Students: " + f);
        System.out.println("Average marks (all students): " +
(sum/studentCount));
    }
}

public class task5 {
    public static void main(String[] args){
        Student s1 = new Student(123, "Ali", 70.2);
        Student s2 = new Student(456, "Ahsan", 80);
        Student s3 = new Student(789, "Owais", 39.33);
        Student s4 = new Student(101, "Umer", 66);
        Student s5 = new Student(321, "Daniyal", 34.0);

        Student[] studentArray = {s1, s2, s3, s4, s5};

        College college = new College("FAST-NUCES", "Karachi",
studentArray);
        college.displayCollegeDetails();
    }
}
```

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Output (Task-5(b)):

```
1   class Student {
8           retu n ID;
9       }
10      public String getName() {
11          return name;
```

PROBLEMS 7 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
ID: 789
Name: Owais
Marks: 39.33
==== Student Information ====
ID: 101
Name: Umer
Marks: 66.0
==== Student Information ====
ID: 321
Name: Daniyal
Marks: 34.0

Total Students: 5
Passed Students: 3
Failed Students: 2
Average marks (all students): 57.00599999999999
PS C:\Users\Administrator\Documents\Second Semester\OOP LAB\La
```